# SMART POLYPHARMACY AND VITAMIN RECOMMENDATION SYSTEM FOR GERIATRIC PATIENTS

25-26J-368

Project Proposal Report

Tharaka B.W.P – IT22600516

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
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August 2025

# PERSONALIZED MENTAL HEALTH AND LIFESTYLE ADVISORY SYSTEM FOR GERIATRIC PATIENTS

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Tharaka B.W.P – IT22600516

Supervised by Ms. Jenny Krishara

Co-supervised by Dr. Dinuka R Wijendra

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# DECLARATION

## DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name | Student ID | Signature |
|---|---|---|
| Tharaka B.W.P | IT22600516 | Tharaka |

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Name of supervisor: Ms. Jenny Krishara

Name of co-supervisor: Dr. Dinuka R Wijendra

Signature of the Supervisor ..... Jenny ..... Date ..... 2025/08/29 .....

Signature of the Co-Supervisor ..... Date ..... 2025/08/29 .....

# ABSTRACT

The AI Lifestyle & Mental Wellness Advisor is an intelligent system designed to support geriatric patients, especially those managing multiple medications, by providing non-pharmacological and non-medical daily guidance for stress reduction, emotional stability, and healthy living. Unlike existing wellness apps that focus solely on reminders or general advice, this system integrates polypharmacy risk scores, patient-reported mental health data, and facial expression analysis to generate personalized, risk-aware recommendations.

The platform delivers practical, drug-free interventions such as mindfulness exercises, relaxation techniques, proper sleep hygiene, hydration tracking, and safe physical activity, tailored to the patient's overall health and medication burden. It also performs continuous monitoring to identify warning signs of severe distress—such as suicidal thoughts, depression, or anxiety spikes—and immediately directs the patient to seek professional medical assistance. Through motivational check-ins, emotional support, and lifestyle coaching, the advisor promotes independence, safety, and well-being among elderly users.

This research is innovative because it is the first system to fuse polypharmacy risk scoring, self-reported mental health status, and facial expression monitoring into a unified AI companion for geriatric care. The approach employs AI-based personalization, lightweight interfaces for accessibility, and multi-modal mental health indicators for accurate risk assessment. Expected outcomes include improved mood, reduced stress, healthier daily routines, and enhanced quality of life for elderly patients managing complex medication regimens.



**Keywords: Geriatric care, Polypharmacy, AI companion, Mental wellness, Lifestyle guidance, Drug-free intervention**

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# LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| AI | Artificial Intelligence |
| SLIIT | Sri Lanka Institute of Information Technology |
| ICT | Information and Communication Technology |
| CBT | Cognitive Behavioral Therapy |
| NLP | Natural Language Processing |
| PSS | Perceived Stress Scale |
| GDS | Geriatric Depression Scale |
| ML | Machine Learning |
| CNN | Convolutional Neural Network |
| API | Application Programming Interface |

Table 1:List of abbreviations

# 1. INTRODUCTION

## 1.1 Background

The world is experiencing a significant demographic shift, with the number of people aged 60 and above projected to nearly double to 2 billion by 2050 [1]. Aging is associated with multiple chronic conditions such as diabetes, cardiovascular diseases, arthritis, and dementia, which often require polypharmacy—the concurrent use of five or more medications [2]. While necessary, polypharmacy increases the risk of adverse drug reactions (ADRs), drug–drug interactions, poor adherence, and frequent hospitalizations [3].

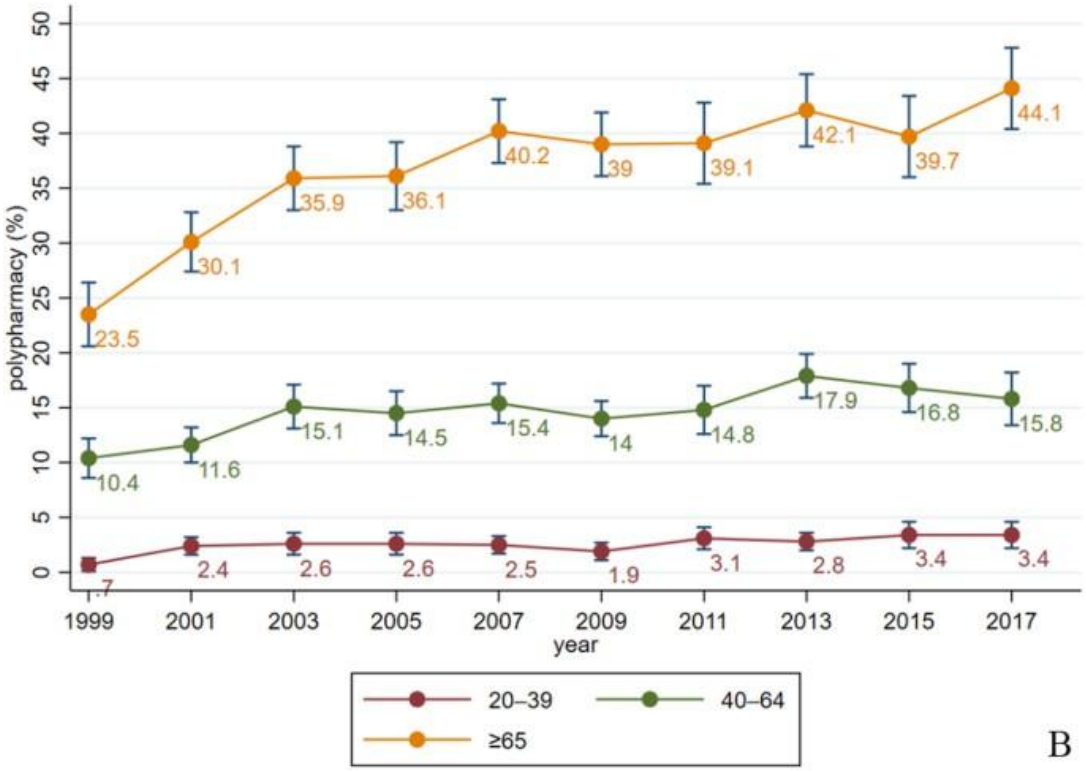


Figure 1:Prevalence and trend of polypharmacy by age [13]

Beyond physical risks, geriatric patients face under-recognized mental health challenges. Studies show that 15–20% of older adults suffer from anxiety or depression, and nearly 50% experience sleep disorders [4]. These problems often worsen physical illness, reduce

independence, and increase cognitive decline. Yet, many elderly individuals suffer in silence due to stigma, lack of awareness, or inadequate geriatric mental health services.

Digital health tools have emerged to assist older patients, but most focus on medication compliance and symptom tracking rather than holistic lifestyle and mental wellness. Very few integrate emotional well-being with drug safety, leaving a critical gap in comprehensive care [5].

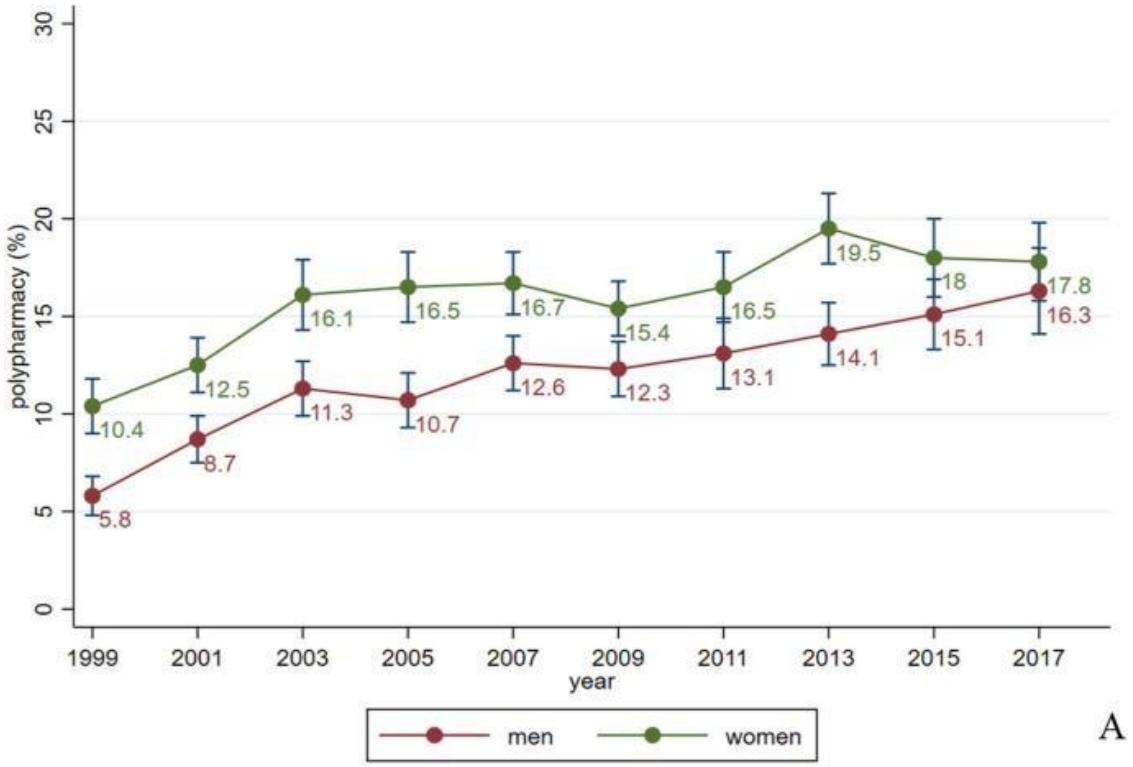


Figure 2:Prevalence and trend of polypharmacy by sex [13]

### 1.2 Literature Review

Research has consistently shown that non-pharmacological interventions are effective for managing late-life stress, anxiety, and sleep issues.

- Mindfulness and relaxation practices improve sleep quality and reduce anxiety, with lasting effects up to six months [6].
- Cognitive Behavioral Therapy (CBT) for insomnia has been shown to significantly reduce both sleep disturbances and depressive symptoms in older adults.
- Lifestyle-based practices such as journaling, physical activity, hydration, and maintaining proper sleep hygiene are safer and more sustainable than drug interventions in elderly populations.

At the same time, AI-driven digital companions are being explored in geriatric care. Many provide functions such as facial expression analysis for stress detection or reminders for medication adherence. However, most are generalized mental health apps that do not address the unique risks of polypharmacy or provide personalized, risk-aware daily guidance.

This gap suggests the need for a system that goes beyond drug reminders, combining polypharmacy risk awareness, mental health monitoring, and lifestyle optimization into a single AI-based digital companion tailored for elderly patients.

### 1.3 Research Gap

Despite significant progress in digital health technologies, current solutions for elderly care remain primarily focused on medication adherence and prescription management. Most platforms provide features such as dosage tracking, reminders, and interaction alerts, but they fail to address the interconnected role of mental health, emotional wellbeing, and lifestyle behaviors in overall treatment success. This leaves a critical gap in providing holistic care for geriatric patients who are managing both polypharmacy and age-related psychological challenges.

**Limitations of Existing Digital Companions**

The majority of existing healthcare apps function as reminder tools, notifying patients when to take their medicines or refill prescriptions. While useful, these systems are medication-centric and lack features that promote daily motivation, emotional support, or stress management. As a result, elderly patients are often left without tools that encourage resilience, healthy lifestyle practices, or mental wellness—factors proven to directly impact treatment adherence and health outcomes [5].

### Unmet Mental Health and Lifestyle Needs

Elderly populations experience a rising burden of stress, anxiety, depression, and sleep disturbances, often linked to chronic illness, cognitive decline, and social isolation [6]. These conditions worsen physical health outcomes and increase the risks associated with polypharmacy [7]. Yet, current polypharmacy platforms do not include preventive mental health support, nor do they adapt guidance based on stress levels, mood changes, or lifestyle factors [8].

### Lack of Personalized, Drug-Free Interventions

Evidence strongly supports non-pharmacological strategies—such as mindfulness, breathing exercises, journaling, and improved sleep hygiene—for improving mental health in older adults. These approaches reduce reliance on additional drugs for insomnia or anxiety, thereby mitigating the risks of polypharmacy. However, most current digital health tools provide only generic advice, without personalization based on a patient's risk profile, emotional state, or daily routine.

### Need for Motivational and Context-Aware Support

Feelings of loneliness and isolation are common in older adults, especially those living alone or in assisted care. Current apps are static and transactional, offering reminders but not meaningful engagement. There is an urgent need for an AI-driven system that can provide daily motivational check-ins, personalized encouragement, and adaptive guidance based on patient-reported data, facial expressions, and polypharmacy risk scores. Furthermore, few systems have built-in safety mechanisms to detect severe mental distress (e.g., suicidal ideation) and recommend urgent professional intervention.

### Towards a Holistic Advisory Companion

The identified gap highlights the absence of a comprehensive AI-based companion that integrates polypharmacy risk awareness with personalized mental health and lifestyle support. Such a system should deliver:

- **Drug-free lifestyle recommendations** (stress reduction, sleep hygiene, safe physical activity, hydration).
- **Personalized mental wellness guidance** (mindfulness, relaxation, journaling, daily motivation).
- **Emotion-aware interactions** (using NLP and affective computing for empathetic support).

- **Safety protocols for severe distress**, ensuring timely referral to professionals.

By addressing these gaps, the proposed system shifts elderly care from a medication-centered model toward a holistic wellness paradigm, supporting both mental and physical health.

### 1.4 Research Problem

The growing dependence of elderly patients on polypharmacy highlights the urgent need for innovative solutions that not only support medication management but also address the broader dimensions of mental wellness and lifestyle care. Traditional healthcare applications and reminder-based systems tend to focus narrowly on timely medication adherence, overlooking the complex interplay between daily lifestyle choices, emotional health, and overall treatment outcomes. This creates a critical gap in elderly care, where patients may take medications correctly yet continue to suffer from untreated stress, depression, or lifestyle-related issues that undermine both their physical and mental well-being.

**Limitations of Current Approaches**

Most existing digital health solutions are medication-centric, designed to provide reminders, track prescription histories, and notify caregivers of missed doses. While useful for adherence, these platforms rarely incorporate non-pharmacological interventions such as mindfulness, relaxation techniques, or lifestyle guidance. As a result, they fail to address daily emotional distress, fatigue, anxiety, or motivational challenges that strongly influence treatment outcomes in geriatric care.

**The Dual Challenge – Physical and Mental Wellness**

Elderly patients face the dual burden of managing chronic physical conditions with multiple medications while also coping with mental health challenges such as stress, anxiety, depression, and social isolation. These psychological difficulties often lead to poor adherence and a vicious cycle, where declining mental health worsens physical conditions, and the increasing complexity of polypharmacy heightens psychological strain. Without integrated support, patients risk losing independence, motivation, and quality of life.

**Need for a Personalized AI-Driven Companion**

Current wellness applications generally rely on static, one-size-fits-all recommendations that do not account for daily stress levels, mood fluctuations, or polypharmacy-related

risks. In contrast, elderly patients require dynamic, adaptive support. An AI-powered system using Natural Language Processing (NLP), affective computing, and behavioral analytics could identify emotional states, monitor stress and mood, and provide real-time, personalized interventions such as breathing exercises, sleep hygiene tips, or motivational coaching. Importantly, the system could also integrate polypharmacy risk scores to adapt lifestyle guidance in alignment with patients' medication burden.

**Balancing Medical Care and Drug-Free Interventions**

A major challenge is ensuring that AI-driven lifestyle and mental health support complements rather than replaces medical treatment. The system must act as a drug-free advisory companion, offering stress reduction, mental wellness exercises, and lifestyle optimization without contradicting or interfering with physician-prescribed care. This adjunctive role can strengthen resilience, improve adherence, and reduce reliance on unnecessary medications for secondary conditions such as anxiety or insomnia.

**Addressing Ethical, Safety, and Adoption Concerns**

Given the sensitivity of mental health data, the system must ensure strict privacy, security, and ethical safeguards. Over-reliance on AI is a concern, as elderly users may mistake digital advice for medical instruction. Thus, clear boundaries must distinguish supportive lifestyle guidance from professional medical care. Furthermore, adoption challenges must be addressed: the system should offer a simple, voice-enabled, and user-friendly interface tailored to older adults with limited digital literacy.

**Defining the Research Problem**

Considering these challenges, the core research problem can be defined as follows:

How can an AI-powered companion be designed to provide elderly patients with personalized, drug-free lifestyle and mental health guidance—integrating polypharmacy risk scores, emotional monitoring, and behavioral insights—while complementing rather than replacing medical treatment?

To address this, the proposed solution must incorporate behavioral monitoring, emotional recognition, adaptive lifestyle interventions, motivational engagement, and safety mechanisms within a holistic AI-driven advisory system. By doing so, it will create a comprehensive care ecosystem that empowers elderly patients, improves treatment compliance, and enhances both mental and physical health outcomes.

## 2. OBJECTIVES

The main goal of this research is to design and develop an AI-powered Lifestyle & Mental Wellness Advisory System that provides drug-free, non-medical guidance for elderly patients, particularly those under polypharmacy care. Unlike conventional reminder systems, this solution integrates polypharmacy risk scores, emotional state recognition, and behavioral monitoring to deliver personalized lifestyle advice, stress-reduction strategies, and motivational support. The system is intended to act as a supportive digital companion that improves overall wellbeing while ensuring that any serious medical or psychological issues are referred to healthcare professionals.

### 2.1 Main Objective

The primary objective of this research is to create an AI-based advisory companion that enhances mental wellness and lifestyle management for geriatric patients on multiple medications. The system will use polypharmacy risk scoring, affective computing, and patient-reported data to provide tailored lifestyle guidance, emotional support, and early detection of mental distress. Its purpose is to **c**omplement existing medical treatment, helping elderly patients maintain a healthier daily routine, reduce stress, and improve quality of life.

### 2.2 Specific Objectives

1. **Integrate polypharmacy risk scoring into lifestyle guidance**
   Use Jayalath's polypharmacy risk score as an input factor to personalize the intensity and type of lifestyle and wellness advice provided to patients.

2. **Collect and analyze patient-reported emotional and lifestyle data**
   Develop a framework to record and analyze data on stress levels, mood, sleep quality, physical activity, and daily habits to create a holistic view of the patient's wellbeing.

3. **Provide non-pharmacological, personalized wellness recommendations**
   Deliver AI-driven advice such as mindfulness practices, breathing exercises, relaxation techniques, physical activity prompts, and sleep hygiene guidance to promote mental balance and healthy living.

4. **Offer daily motivational and lifestyle check-ins**
   Implement an interactive digital companion that provides motivational prompts, positive affirmations, and adaptive reminders to help patients stay engaged and feel emotionally supported.

5. **Detect and respond to signs of severe mental distress**
   Incorporate AI-based monitoring to identify high-risk patterns such as hopelessness, withdrawal, or suicidal thoughts. When detected, the system will advise immediate professional consultation while maintaining patient safety.

# 3. METHODOLOGY

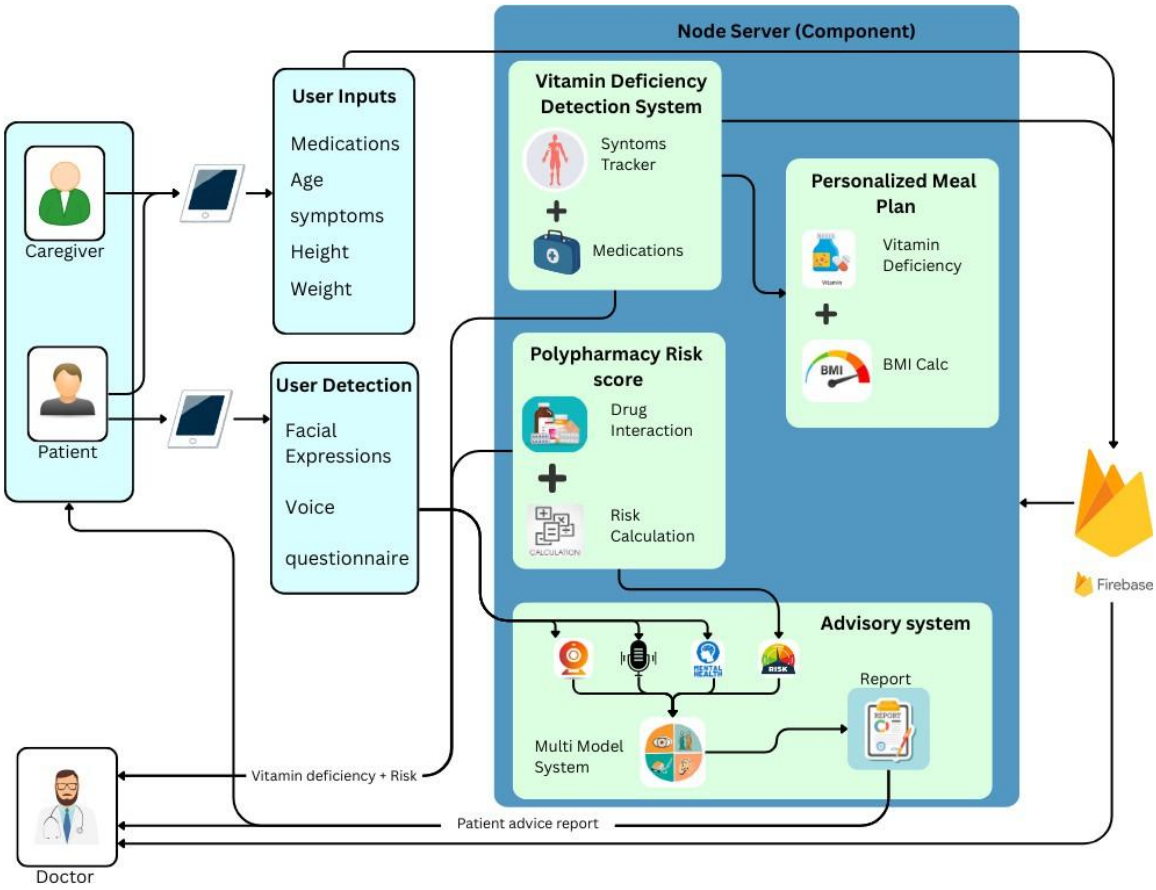


Figure 3:Overall System Overview Diagram

This study proposes the design and development of an AI-powered Lifestyle & Mental Wellness Advisory System for geriatric patients. The system integrates polypharmacy risk scores, facial expression recognition, and patient-reported data to deliver personalized, non-pharmacological lifestyle and mental wellness guidance. Unlike existing polypharmacy-focused applications, this approach provides a holistic, adaptive companion that addresses both the physical and psychological dimensions of elderly care.

## 3.1 Data

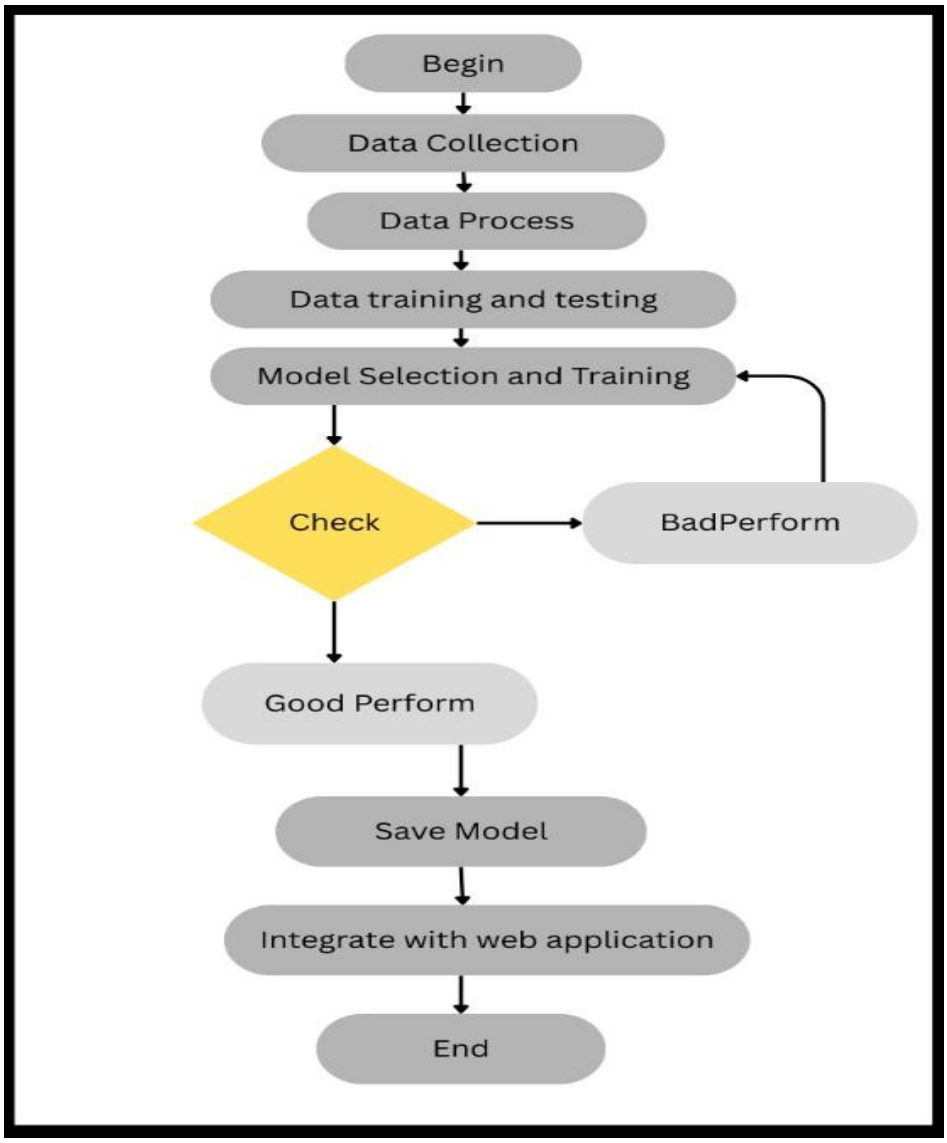


Figure 4:Machine Learning Training Process for Mental Health Detection

**Collection and Inputs**

The system will collect multimodal patient data, including:

- **Self-reported information**: daily mood, stress levels, sleep quality, and activity logs (via mobile or web interface).
- **Facial expression analysis**: AI-driven emotion recognition to assess real-time mood states.

- **Polypharmacy risk score**: integrated from the medication-based risk scoring system, representing medication complexity and vulnerability.
- **Behavioral data**: frequency of engagement with the system, journaling responses, and adherence to wellness suggestions.

This ensures both **objective (facial/emotional)** and **subjective (self-reported)** inputs are captured.

### 3.2 Personalized Lifestyle Guidance Using AI Algorithms

Machine learning models will process the collected data to generate dynamic, individualized lifestyle advice:

- **Sleep irregularities** → recommendations for sleep hygiene (consistent bedtime, reducing screen time).
- **Low activity levels** → light, elderly-friendly exercise prompts.
- **Hydration or dietary inconsistencies** → reminders and nutrition balance tips.
- **High polypharmacy risk scores** → stricter monitoring and more frequent motivational prompts.

The system adapts over time using reinforcement learning, ensuring advice becomes more context-aware and personalized as patient interaction increases.

### 3.3 Mental Wellness Interventions

The platform provides drug-free, evidence-based techniques for emotional support:

- Mindfulness exercises (guided breathing, meditation).
- Journaling prompts (reflection, gratitude, emotional expression).
- Relaxation techniques (progressive muscle relaxation, calming audio).
- Motivational check-ins (daily affirmations, empathetic engagement).

Recommendations will be dynamically adjusted based on facial expression analysis and self-reported stress/mood scores.

### 3.4 Risk Detection Using AI and Psychological Scoring Metrics

To ensure early detection of mental distress, the system applies:

- **Polypharmacy risk score integration** → higher medication risks linked to greater emotional burden.

- **Facial emotion recognition** → signs of stress, sadness, or withdrawal.
- **Validated psychological scales** → Perceived Stress Scale (PSS), Geriatric Depression Scale (GDS).
- **Predictive ML classifiers** → detect potential red flags like severe depression or suicidal ideation.

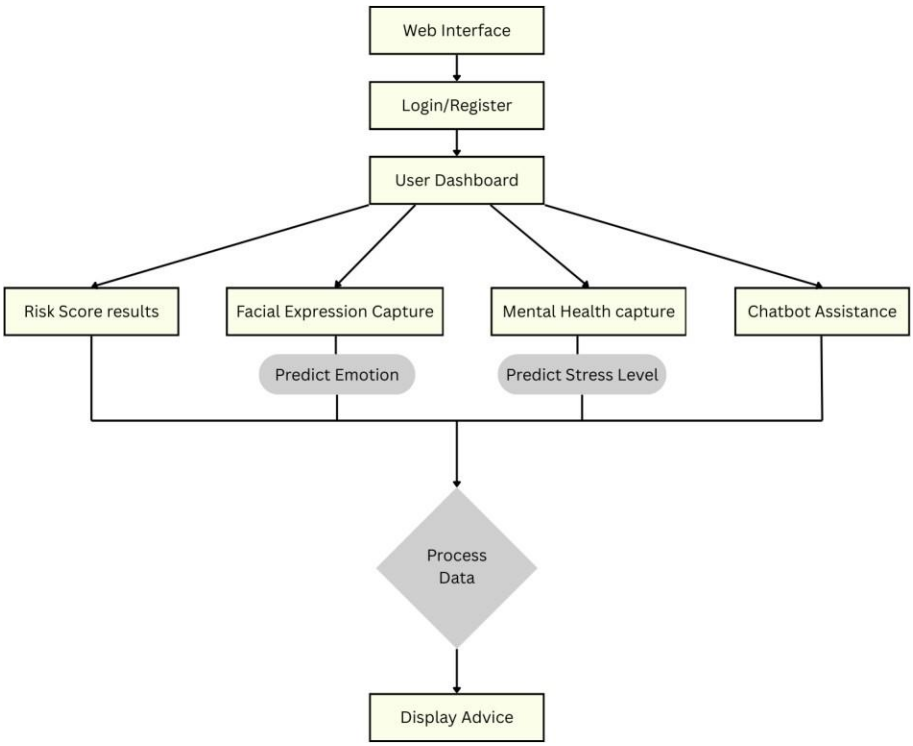


Figure 5:Flowchart of the Application

### 3.5 Safety Protocols and Escalation Measures

The system includes automated safeguards to protect patients:

1. **Immediate alerts** when severe distress is detected, advising professional care.
2. **Emergency referrals** with contact details for caregivers and professionals.
3. **Optional caregiver notifications** (with patient consent) in critical cases.
4. **Clear system boundaries** – AI never prescribes medication, only complements medical treatment.

### 3.6 System Architecture and Workflow

The workflow is designed as follows:

1. **Input Layer** – patient-reported data, facial expressions, polypharmacy risk score.
2. **Processing Layer** – ML models analyze lifestyle trends, emotional states, and medication-related risk levels.
3. **Output Layer** –
    - Personalized lifestyle recommendations.
    - Mental wellness interventions (mindfulness, journaling, relaxation).
    - Daily motivational prompts.
    - Risk alerts and professional care suggestions.
4. **Feedback Loop** – adaptive learning updates patient profiles continuously, refining advice with new inputs.

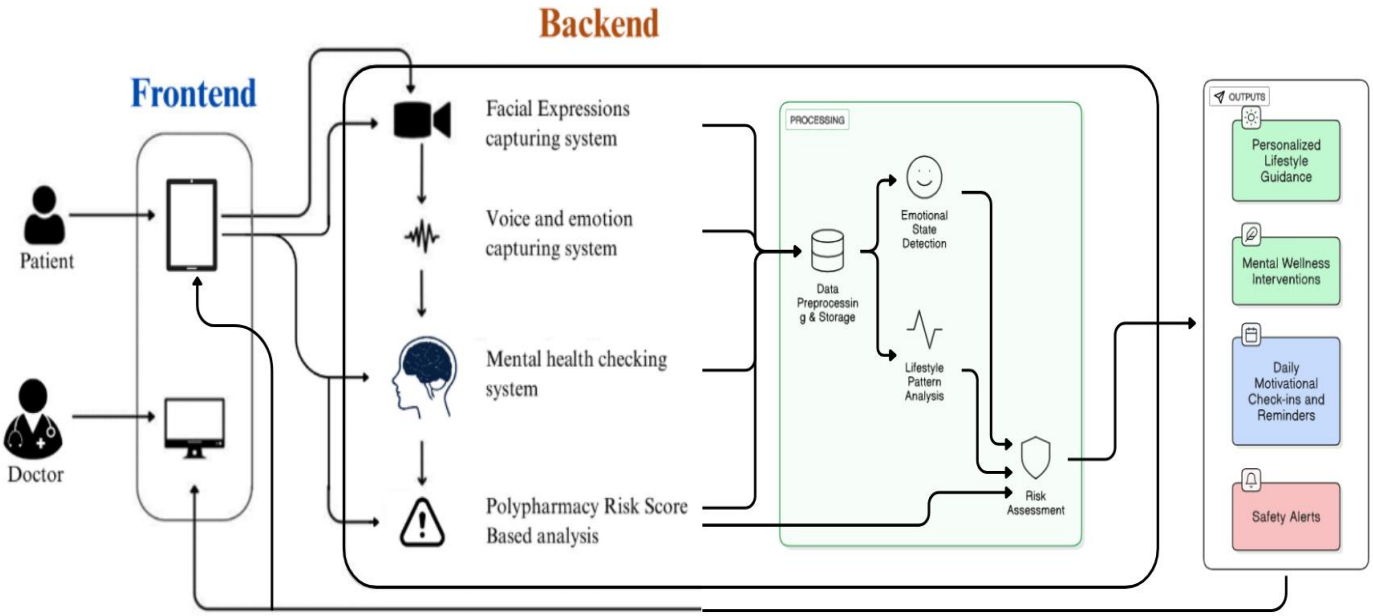


Figure 6:System diagram of Component

## 4. DESCRIPTION OF PERSONAL AND FACILITIES

Developing an AI-based Lifestyle & Mental Wellness Companion for elderly patients requires a careful balance of functional, user, system, and non-functional requirements. These ensure that the platform is technically sound, clinically safe, and user-friendly for older adults.

### 4.1 Functional Requirements

Functional requirements define the core features the system must deliver:

- **Patient-Reported Data Collection**: Patients can log daily routines, stress levels, sleep quality, mood, and activity patterns via structured checklists, sliders, or voice input. Voice-based interaction is prioritized to assist seniors with limited digital literacy.
- **Facial Expression Recognition**: The AI system must analyze facial expressions in real-time to detect mood states such as stress, sadness, or low engagement, complementing self-reported data.
- **Personalized Lifestyle Guidance**: Using machine learning, the system must generate lifestyle advice tailored to each user's health profile, polypharmacy risk score, and emotional state (e.g., sleep hygiene, hydration reminders, gentle exercises).
- **Mental Wellness Recommendations**: The platform should provide non-pharmacological interventions like mindfulness exercises, journaling prompts, relaxation practices, and breathing routines, dynamically adjusted based on mood/stress input.
- **Daily Motivational Check-ins**: The companion must deliver empathetic reminders, positive affirmations, and motivational interactions to sustain patient engagement and reduce loneliness.
- **Risk Detection and Safety Alerts**: AI models must flag warning signs of severe mental distress (e.g., suicidal ideation, acute depression) using psychological scoring and emotional signals. With prior consent, alerts may also notify caregivers or family members.

### 4.2 User Requirements

User requirements ensure the system is **accessible, safe, and supportive**:

- **Elderly Patients**:
  - A simple, minimalistic interface with large fonts, clear icons, and voice-based input/output.
  - Accessible across different literacy levels and adaptable for cognitive decline.
  - Privacy-preserving, ensuring sensitive lifestyle and emotional data remain confidential.
- **Caregivers and Family**:
  - With patient consent, authorized caregivers can access summary reports and distress alerts.
  - Ability to receive real-time notifications in emergencies.
- **Cross-Platform Accessibility**:
  - Application must work seamlessly on smartphones, tablets, and desktops.
  - Offline features (local caching) for patients with inconsistent internet access.

### 4.3 System Requirements

System requirements define the technical specifications needed to implement and run the application effectively.

- **Hardware Requirements**:
  - **Patients**: Laptops or tablets with front-facing cameras (for facial recognition), stable internet, and basic audio capabilities.
  - **Caregivers or Supervisors**: Desktop computers or tablets with sufficient processing power to view reports, alerts, and dashboards.
- **Software Requirements**:
  - **Operating Systems**: Windows and iOS support.
  - **Programming Languages**: Python for AI and machine learning, Flask or FastAPI for API development, and React/Next.js for front-end applications.

- **Data Processing Libraries**: Libraries such as Pandas for data management, NumPy for numerical analysis, and Librosa for optional audio processing if voice inputs are used.
- **Machine Learning Frameworks**: TensorFlow or PyTorch for training and deploying models for stress detection and lifestyle recommendations.
- **Containerization**: Docker to ensure consistent deployment across different platforms.
- **Database**: Secure cloud database (e.g., Firebase, PostgreSQL) for storing patient-reported data, recommendations, and engagement logs.

### 4.4 Non-Functional Requirements

Non-functional requirements describe the qualities and constraints of the system, focusing on performance, reliability, and safety.

- **Accuracy**: The AI system must provide highly accurate recommendations and risk assessments based on validated models and patient-reported data [8].
- **Real-Time Responsiveness**: Recommendations, motivational check-ins, and risk alerts should be delivered in real-time, allowing immediate action when necessary [9].
- **Usability**: The interface must be intuitive and simple, allowing elderly patients to navigate the system without extensive training. Graphical indicators, easy-to-read fonts, and minimalistic design should be used.
- **Security and Privacy**: All patient data must be encrypted during transmission and storage. Role-based access control must be implemented to prevent unauthorized access [10].
- **Scalability**: The system must support multiple concurrent users, handling long-term data storage and analysis without degradation in performance.
- **Compliance**: The system must comply with data protection regulations such as GDPR, HIPAA, or local healthcare privacy standards, ensuring ethical handling of sensitive patient data.
- **Reliability and Availability**: The system should be operational 24/7, with minimal downtime, ensuring that patients can access support whenever needed.

In conclusion, fulfilling these thorough requirements is essential to the project's success. Non-functional requirements guarantee efficiency, security, and

dependability; user requirements concentrate on accessibility and ease of use; system requirements offer the technological framework; and functional requirements guarantee that the system accomplishes the required duties efficiently. The AI Lifestyle & Mental Wellness Companion can provide older patients with comprehensive care by meeting all of these requirements, enhancing mental and emotional health and assisting with medical therapy.

# 5. BUDGET AND BUDGET JUSTIFICATION

| **Item** | **Estimated Cost (LKR)** |
|---|---|
| Software and Development Tools | 12,000 |
| Cloud Hosting & Deployment (monthly) | 19,999 |
| Web App Hosting (monthly) | 10,000 |
| Database (monthly) | 7,000 |
| Monitoring & Security | 9,000 |

Table 2:Budget table

## 5.1 Budget Justification

**Software and Development Tools**

Software and development tools such as Android Studio, Xcode, and Visual Studio Code are essential for building and testing the cross-platform mobile application. Additional accessibility testing libraries, Text to-Speech plugins, and gesture detection frameworks may require licensing or premium versions to ensure compatibility and high performance. These tools form the foundation of the development process and are necessary to design, implement, and refine the gesture–vibration interface.

**Hosting & Cloud Services**

Hosting and cloud services are required to store and synchronize educational content, quiz data, and user progress between the mobile app and the teacher dashboard. Cloud infrastructure such as Firebase, AWS, or Google Cloud will also enable analytics, backup, and secure user data management. Reliable hosting ensures that visually impaired learners can access lessons both online and offline while maintaining data privacy and continuity.

**Google Play Store Publishing Fee (One-Time)**

A one-time Google Play Store developer registration fee is mandatory to distribute the application on Android devices. This fee ensures official app deployment, automatic updates, and visibility to global users. Publishing through the Play Store also allows compliance with security and accessibility policies, ensuring the app reaches visually impaired learners in a trustworthy and reliable manner.

**Apple App Store Developer Fee (Annual)**

The Apple App Store requires an annual developer fee for publishing applications on iOS devices. This cost is essential to make the application available to iPhone and iPad users, ensuring inclusivity across platforms. The annual subscription also provides access to Apple's developer tools, beta testing environments, and app analytics, which are necessary for maintaining quality and accessibility compliance.

## 6. GANTT CHART

| No | Assessment (Milestones) | 2025 - 2026 | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | May (25) | Jun (25) | Jul (25) | Aug (25) | Sep (25) | Oct (25) | Nov (25) | Dec (25) | Jan (26) | Feb (26) | Mar (26) | Apr (26) | May (26) |
| 1 | Project Discussion Workshop 1 | | | | | | | | | | | | | |
| 1.1 | Project Discussion Workshop 2 | | | | | | | | | | | | | |
| 2 | Topic Evaluation | | | | | | | | | | | | | |
| 2.1 | Select a Topic | | | | | | | | | | | | | |
| 2.2 | Select a supervisor | | | | | | | | | | | | | |
| 2.3 | Topic Evaluation Form Submission | | | | | | | | | | | | | |
| 3 | Project Proposal Report | | | | | | | | | | | | | |
| 3.1 | Create Project Proposal Report (draft) | | | | | | | | | | | | | |
| 3.2 | Create Project Proposal Presentation | | | | | | | | | | | | | |
| 3.3 | Project Proposal Presentation | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3.4 | Create Project Proposal (Individual) | | | | | | | | | | | | | |
| 3.5 | Create Project Proposal (Group) | | | | | | | | | | | | | |
| 4 | Develop the System | | | | | | | | | | | | | |
| 4.1 | Identifying Functions | | | | | | | | | | | | | |
| 4.2 | Database Designing | | | | | | | | | | | | | |
| 4.3 | Implementation | | | | | | | | | | | | | |
| 4.4 | Unit testing | | | | | | | | | | | | | |
| 4.5 | Integration Testing | | | | | | | | | | | | | |
| 5 | Progress Presentation - I | | | | | | | | | | | | | |
| 5.1 | Project Status document | | | | | | | | | | | | | |
| 5.2 | Create Presentation Document | | | | | | | | | | | | | |
| 5.3 | Progress Presentation – I (20%) | | | | | | | | | | | | | |
| 6 | Research Paper | | | | | | | | | | | | | |
| 6.1 | Create the Research Paper | | | | | | | | | | | | | |
| 6.2 | Final Research paper | | | | | | | | | | | | | |
| 7 | Progress Presentation - II | | | | | | | | | | | | | |
| 7.1 | Create Presentation Document | | | | | | | | | | | | | |
| 7.2 | Progress presentation – II (80%) | | | | | | | | | | | | | |
| 8 | Final Report Submission | | | | | | | | | | | | | |
| 8.1 | Final Report Submission | | | | | | | | | | | | | |
| 8.2 | Application Assessment | | | | | | | | | | | | | |
| 8.3 | Project Status Document | | | | | | | | | | | | | |
| 8.4 | Student Logbook | | | | | | | | | | | | | |
| 9 | Final Presentation & Viva | | | | | | | | | | | | | |
| 9.1 | Create Final Presentation | | | | | | | | | | | | | |
| 9.2 | Final Report Submission | | | | | | | | | | | | | |

Table 3:gantt chart

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